

Chapter 1

TT

1.1 SYS-0

The alphabet is listed as follows.

- ◇ $V = \{v_i\}_{i \geq 0} \cup \{A_i\}_{i \geq 0}$;
- ◇ C is the set of constant symbols;
- ◇ $B = \{\Pi, \lambda, \Sigma, \dots\}$ is the set of binders;
- ◇ $U = \{\text{Type}_i\}_{i \geq 0}$ is the set of universes;
- ◇ $P = \{(\cdot), \cdot, \cdot, \equiv\}$ is the set of punctuation.

All of these sets are at most countable.

The whole formulas we use is contained by set

$$F = V|C|U|B(V : F)F|(FF)|(CF^*).$$

Notation 1.1.1 *Here we announce some commonly used notations for avoiding ambiguity.*

- ◇ $\mathbf{a} = a_0 a_1 \dots a_m$;
- ◇ $f(\mathbf{x}) := f$ where $f \in F$ and $\text{Fr } f \subset \{x_0, \dots, x_m\}$;
- ◇ $(f\mathbf{x})$ refers $(\dots((fx_0)x_1)\dots)$, note that this may differ the notation $(fx_0x_1\dots x_m)$, which is equal to our first definition;
- ◇ for any $b \in B$, $b(\mathbf{v} : \mathbf{A})N := b(v_0 : A_0)b(v_1 : A_1)\dots b(v_m : A_m)N$;
- ◇ $\mathbf{v} : \mathbf{A}$ refers

$$v_0 : A_0 \quad v_1 : A_1 \quad \dots \quad v_s : A_s;$$
- ◇ $M \rightarrow N := \Pi(v : M)N$, in which $v \notin \text{Fr } N$;
- ◇ to distinguish between terms in F and constants in C , we use a mono font for constants;
- ◇ $\mathcal{U} := \text{Type}_{\aleph_0}$, and $M : \mathcal{U}$ implies there exists a $k < \aleph_0$ such that $M : \text{Type}_k$.
- ◇ for any relation $\mathcal{R} \subset F^2$, $\text{Compatible}\mathcal{R}$ is a abbreviation of the statement: for all if MRN , then for any $L, A, \phi_1, \dots, \phi_s \in F$, $b \in B$, $\mathbf{c} \in C$ and $v \in V$, $(ML)\mathcal{R}(NL)$, $(LM)\mathcal{R}(LN)$, $b(v : A)MRb(v : A)N$ and

$$(\mathbf{c}\phi_1 \dots \phi_k M \phi_{k+1} \dots \phi_s)\mathcal{R}(\mathbf{c}\phi_1 \dots \phi_k N \phi_{k+1} \dots \phi_s)$$

Definition 1.1.2 A definition D has the form

$$\mathbf{c}(v : \mathbf{A}) \equiv d(v) : B,$$

where $v \in V$, $\mathbf{c} \in C$ and all of $A, B, d(v)$ are in F .

Definition 1.1.3 A sequent has the form

$$D \quad v : \mathbf{A} \quad M : A$$

where $M : A$ is called a judgement, and both M and A are in F . The set of all sequents is denoted by Sqt .

Legal judgements consist a subset of $F : F$, which is derived from a series of **rules**. In our SYS-0, we first introduce the following rules, each of them can be viewed as a relation on $\text{Sqt}^* \times \text{Sqt}$.

Sort

If $m < n$, then

$$_ _ \text{Type}_m : \text{Type}_n$$

Var

If $v \in V \setminus (\Gamma \cap V)$, then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad v : A \end{array}$$

Weak

If $v \in V \setminus (\Gamma \cap V)$, then

$$\begin{array}{l} D \quad \Gamma \quad M : A \\ D \quad \Gamma \quad B : \mathcal{U} \\ D \quad \Gamma \quad v : B \quad M : A \end{array}$$

Conv

For an arbitrary equivalence relation $=_a \subset F^2$, if $B =_a B'$, then

$$\begin{array}{l} D \quad \Gamma \quad M : B \\ D \quad \Gamma \quad B' : \mathcal{U} \\ D \quad \Gamma \quad M : B' \end{array}$$

Def

If $D = \mathbf{c}(v : \mathbf{A}) \equiv d(v) : B$, in which $d(v)$ can be an empty string $_$, then

$$\begin{array}{l} D \quad \Gamma \quad K : L \\ D \quad \Gamma \quad v : \mathbf{A} \quad M : N \\ DD \quad \Gamma \quad K : L \end{array}$$

Inst

If $\mathbf{c}(v : \mathbf{A}) \equiv d(v) : B \in D$, where d may be an empty string, then

$$\begin{array}{l} D \quad \Gamma \quad \text{Type}_0 : \text{Type}_1 \\ D \quad \Gamma \quad \xi_0 : A \frac{\xi_0 \cdots \xi_m}{v_0 \cdots v_m} \\ \vdots \\ D \quad \Gamma \quad \xi_m : A \frac{\xi_0 \cdots \xi_m}{v_0 \cdots v_m} \\ D \quad \Gamma \quad (\mathbf{c}\xi) : B \frac{\xi_0 \cdots \xi_m}{v_0 \cdots v_m} \end{array}$$

Compu

Definition 1.1.4 For any definition D , $\rightarrow_D$ is the samllest relation constants

$$(c\xi) \rightarrow_D d(\xi),$$

for all $c \in D$, $\xi \in F^*$, and satisfies Compatible $\rightarrow_D$.

1.2 SYS-1

我们通过引入新符号, 增加新规则和关系来增加新的项和类型, 这个过程大致遵循一个固定的工作流: 1. Formation: 引入新规则, 让系统在指定条件下可以导出新类型; 2. Constuction: 引入新规则, 让系统在指定条件下可以导出新项, 并且这些项都具有新类型; 3. Elimination: 引入新规则, 告知系统新项应该如何使用; 4. Computaiton: 引入 F 上的新关系, 告知系统新系统中某些表达和旧系统中的某些表达具有同样的语义.

现在我们将拓展 SYS-0, 增加 λ -Terms 和 Π -Types.

SYS-1: Form

If $s_1 \leq s_2$, then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad B : \mathcal{U} \\ D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \end{array}$$

SYS-1: Constr

$$\begin{array}{l} D \quad \Gamma \quad v : A \quad M : B \\ D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \\ D \quad \Gamma \quad \lambda(v : A)M : \Pi(v : A)B \end{array}$$

SYS-1: Elim

$$\begin{array}{l} D \quad \Gamma \quad M : \Pi(v : A)B \\ D \quad \Gamma \quad N : A \\ D \quad \Gamma \quad (MN) : B \frac{N}{v} \end{array}$$

SYS-1: Compu

Definition 1.2.1 $\rightarrow_0 \subset F^2$ is the samllest relation satisfies the following properties:

◇ for any A, M in F , $b \in B$ and $v_0, v_1 \in V$ such that $v_1 \in (M \cap V) \setminus \{v_0\}$

$$b(v_0 : A)M \rightarrow_0 b(v_1 : A)M \frac{v_1}{v_0};$$

◇ Compatible $\rightarrow_0$.

Definition 1.2.2 $\rightarrow_1 \subset F^2$ is the samllest relation satisfies the following properties:

◇ for any A, M, N in F , $(\lambda(v : A)MN) \rightarrow_1 M \frac{N}{v}$;

◇ Compatible $\rightarrow_1$.

$=_1$ is the equivalence genertated by $\bigcup \{\rightarrow_i\}_{i=0}^1$.

1.3 SYS-2

在 SYS-1 的基础上加入 Σ -Types 得到 SYS-2.

SYS-2: Form

$$\begin{array}{l}
D \quad \Gamma \quad A : \mathcal{U} \\
D \quad \Gamma \quad v : A \quad B : \mathcal{U} \\
D \quad \Gamma \quad \Sigma(v : A)B
\end{array}$$

SYS-2: Constr

$$\begin{array}{l}
D \quad \Gamma \quad \Sigma(v : A)B \\
D \quad \Gamma \quad a : A \\
D \quad \Gamma \quad b : B_v^a \\
D \quad \Gamma \quad (\text{pair}ab) : \Sigma(v : A)B
\end{array}$$

定义 Elim 前先明确需求, 对任何 Type 或者 Universe Q , 通过给定的项 $\phi : \Pi(v : A)(Bv \rightarrow Q)$, 其中 Bv 指的是公式的合并 (Bv) , 我们可以构造出一个具有类型 $(\Sigma(v : A)Bv) \rightarrow Q$ 的项 ψ . 所以, 有理由将规则设计成下面这个样子:

$$\begin{array}{l}
D \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad \phi : \Pi(v : A)(Bv \rightarrow Q) \\
D \quad \Gamma \quad (\text{Elim}Q\phi) : (\Sigma(v : A)Bv) \rightarrow Q
\end{array}$$

然后再加入新关系 $\rightarrow_2$, 其包含下面内容

$$((\text{Elim}Q\phi)(\text{pair}ab)) \rightarrow_2 ((\phi a)b)$$

除此之外还有一种更普遍的方法, 它是从更复杂的功能中下放到这里的.

SYS-2: Elim

$$\begin{array}{l}
D \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad (\text{Ind}\Sigma(v : A)B) : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow \mathcal{U}). \\
\quad \quad [\Pi(v_1 : A)(v_2 : Bv_2)(X\text{pair}v_1v_2) \rightarrow \Pi(v_3 : \Sigma(v_4 : A)Bv_4)(Xv_3)]
\end{array}$$

可见如果给出常函数 $H = \lambda(X : \Sigma(v_0 : A)B)Q : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow \mathcal{U})$, 还有

$$\phi : \Pi(v_0 : A)(v_1 : Bv_0)H(\text{pair}v_0v_1) = \Pi(v_0 : A)(v_1 : Bv_0)Q,$$

那么

$$(\text{Ind}\Sigma(v : A)B)(H\phi) : (\Sigma(v : A)Bv) \rightarrow Q.$$

SYS-2: Compu

Definition 1.3.1 $\rightarrow_2 \subset F^2$ is the samllest relation satisfies the following properties:

◇ for all $a, b, A, B, \phi, H \in F$,

$$(\text{Ind}\Sigma(v : A)B)(H\phi ab) \rightarrow_2 (\phi a)b$$

◇ Compatible $\rightarrow_2$.

$\Rightarrow_2$ is the equivalence genertated by $\bigcup\{\rightarrow_i\}_{i=0}^2$.

1.4 SYS-3

在 SYS-2 中加入自然数类型 $\mathbb{N}$ 使其成为 SYS-3.

SYS-3: Form

$$\begin{array}{l}
D \quad \Gamma \quad _ \\
D \quad \Gamma \quad \mathbb{N} : \mathcal{U}
\end{array}$$

SYS-3: Constr

$$\begin{array}{l} D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\ D \quad \Gamma \quad \mathbf{0} : \mathbb{N} \end{array}$$

and

$$\begin{array}{l} D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\ D \quad \Gamma \quad \mathbf{S} : \mathbb{N} \rightarrow \mathbb{N} \end{array}$$

SYS-3: Elim

$$\begin{array}{l} D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\ D \quad \Gamma \quad (\mathbf{IndN}) : \Pi(X : \mathbb{N} \rightarrow \mathcal{U})[(X\mathbf{0}) \rightarrow \Pi(v_0 : \mathbb{N})(Xv_0) \rightarrow X(\mathbf{S}v_0)] \rightarrow \Pi(v_1 : \mathbb{N})(Xv_1)] \end{array}$$

Example 1.4.1 Let P be $\lambda(v : \mathbb{N})\text{Type}_0$, which implies that P is a property on $\mathbb{N}$. In addition, assume that p_0 and p_S are two legal elements in $\mathbb{F}$, satisfy $p_0 : P\mathbf{0}$ and $p_S : \Pi(v : \mathbb{N})(Pv) \rightarrow P(\mathbf{S}v)$, means that both propositions $P\mathbf{0}$ and $\forall n : \mathbb{N} Pn \rightarrow P\mathbf{S}n$ are true, then

$$(\mathbf{IndN})(Pp_0p_S) : \Pi(v : \mathbb{N})(Pv),$$

namely, the proposition $\forall n : \mathbb{N} P(n)$ is true.

SYS-3: Compu

Definition 1.4.2 $\rightarrow_3 \subset \mathbb{F}^2$ is the smallest relation satisfies the following properties:

$\diamond$ for all $\phi, a, b, c \in \mathbb{F}$,

$$(\mathbf{IndN})(\phi ab\mathbf{0}) \rightarrow_3 a$$

and

$$(\mathbf{IndN})(\phi ab(\mathbf{S}c)) \rightarrow_3 (bc)((\mathbf{IndN})(\phi abc))$$

$\diamond$ Compatible $\rightarrow_3$.

$=_3$ is the equivalence generated by $\bigcup \{\rightarrow_i\}_{i=0}^3$.

1.5 SYS-CIC

上述流程推广后就得到 SYS-CIC.

Definition 1.5.1 For any $t \in \mathbb{F}$, we have the following definitions:

$\diamond$ an Arity is generated by the syntax

$$\mathbb{A} = U | \Pi(V : \mathbb{F}) \mathbb{A};$$

$\diamond$ Strictly Positive Types of t is the set

$$\mathbb{P}(t) = \{\Pi(v_0 : B_0) \cdots (v_r : B_r)[(\cdots (tN_0) \cdots N_s)] | B_i, N_i \in \mathbb{F} \wedge t \notin B_i, N_i\};$$

in fact, it is a set defined inductively:

$$\begin{aligned} G_0 &= \{t\} \\ G_{n+1} &= \{(xN) | x \in G_n \wedge N \in \mathbb{F} \wedge t \notin N\} \\ H_0 &= \bigcup_{n \geq 0} G_n \\ H_{n+1} &= \{\Pi(v : B)x | B \in \mathbb{F} \wedge x \in H_n \wedge t \notin B\} \\ \mathbb{P} &= \bigcup_{n \geq 0} H_n. \end{aligned}$$

◇ *Forms of Constructor of t is generated by*

$$\begin{aligned}
G_0 &= \{t\} \\
G_{n+1} &= \{(xN) \mid x \in G_n \wedge N \in \mathbf{F} \wedge t \notin N\} \\
H_0 &= \bigcup_{n \geq 0} G_n \\
H_{n+1} &= \{\Pi(v : B)x \mid B \in \mathbf{F} \wedge x \in H_n \wedge [(t \notin B) \vee (t \in B \wedge v \notin x)]\} \\
\mathbb{C} &= \bigcup_{n \geq 0} H_n.
\end{aligned}$$

Definition 1.5.2 *Given $t, Q : \Pi(\mathbf{a} : \mathbf{A})\mathcal{U}$, which is an arity, we can define a function $F_{t|Q}^n : \mathbb{C}(t) \rightarrow \mathbf{F}$ by induction:*

$$\begin{aligned}
F_{t|Q}^n(t) &= Q \\
\text{if } F_{t|Q}^n(x) &\text{ is defined} \\
F_{t|Q}^n(xN) &= F_{t|Q}^n(x)N \\
\text{if } B &\in \mathbf{F} \text{ and } t \notin B, \\
F_{t|Q}^n(\Pi(v : B)x) &= \Pi(v : B)F_{t|Q}^n(x) \\
\text{if } B &\in \mathbb{P}(t) \text{ and } t \in B, \\
F_{t|Q}^n(\Pi(v : B)x) &= B \rightarrow B \stackrel{Q}{t} \rightarrow F_{t|Q}^n(x).
\end{aligned}$$

Definition 1.5.3 *Let $t : \Pi(\mathbf{a} : \mathbf{A})\mathcal{U}$, $Q : \Pi(\mathbf{a} : \mathbf{A})(t\mathbf{a}) \rightarrow \mathcal{U}$, and $c \in \mathbb{C}(t)$, the function $F_{t|Q}^d : \mathbf{F} \times \mathbb{C} \rightarrow \mathbf{F}$ is given by:*

$$\begin{aligned}
F_{t|Q}^d(r, t\mathbf{M}) &= (Q\mathbf{M})r \\
\text{if } F_{t|Q}^d(_, x) &\text{ is defined, for all } r \in \mathbf{F} \\
\text{if } B &\in \mathbf{F} \text{ and } t \notin B, \\
F_{t|Q}^d(r, \Pi(v : B)x) &= \Pi(v : B)F_{t|Q}^d((rv), x) \\
\text{if } \mathbb{P}(t) \ni B &= \Pi(\mathbf{b} : \mathbf{B})(t\mathbf{N}) \text{ and } t \in B, \\
F_{t|Q}^d(r, \Pi(v : B)x) &= \Pi(v : B)[\Pi(\mathbf{b} : \mathbf{B})((Q\mathbf{N})(v\mathbf{b})) \rightarrow F_{t|Q}^d((rv), x)].
\end{aligned}$$

Definition 1.5.4 *Let $t : \Pi(\mathbf{a} : \mathbf{A})\mathcal{U}$, $f \in \mathbf{F}$, the function $F_{t|f}^e : \mathbf{F} \times \mathbb{C}(t) \rightarrow \mathbf{F}$ is given by*

$$\begin{aligned}
F_{t|f}^e(r, t) &= r \\
\text{if } F_{t|f}^e(_, x) &\text{ is defined, for all } r \in \mathbf{F} \\
F_{t|f}^e(r, xN) &= F_{t|f}^e(r, x) \\
\text{if } B &\in \mathbf{F} \text{ and } t \notin B, \\
F_{t|f}^e(r, \Pi(v : B)x) &= \lambda(v : B)F_{t|f}^e((rv), x) \\
\text{if } \mathbb{P}(t) \ni B &= \Pi(\mathbf{b} : \mathbf{B})(t\mathbf{N}) \text{ and } t \in B, \\
F_{t|f}^e(r, \Pi(v : B)x) &= \lambda(v : B)F_{t|f}^e(rv\lambda(\mathbf{b} : \mathbf{B})(f\mathbf{N}(v\mathbf{b})), x).
\end{aligned}$$

CIC: Form

Without losing generality, suppose that $X \in V$, $C_i \in \mathbb{C}(X)$ and $\Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \in \mathbb{A}$, then

$$\begin{array}{lcl}
D & \Gamma & X : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \quad C_0 : \text{Type}_0 \\
& \vdots & \\
D & \Gamma & X : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \quad C_m : \text{Type}_0 \\
D & \Gamma & \lambda : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0
\end{array}$$

CIC: Constr

$$\begin{array}{l} D \quad \Gamma \quad X : \Pi(a : A) \text{Type}_0 \\ D \quad \Gamma \quad (\text{ConstrXi}) : C_i \frac{X}{X} \end{array}$$

CIC: Elim-1

$$\begin{array}{l} D \quad \Gamma \quad \mu : (XW) \\ D \quad \Gamma \quad Q : \Pi(a : A) \text{Type}_0 \\ D \quad \Gamma \quad f_i : [F_{X|Q}^n(C_i)] \frac{X}{X} \\ D \quad \Gamma \quad (\text{IndX})(Qf_0 \cdots f_m \mu) : (QW) \end{array}$$

CIC: Elim-2

$$\begin{array}{l} D \quad \Gamma \quad r_i : C_i \\ D \quad \Gamma \quad \mu : (XW) \\ D \quad \Gamma \quad Q : \Pi(a : A)(Xa) \rightarrow \text{Type}_0 \\ D \quad \Gamma \quad f_i : [F_{X|Q}^d(r_i, C_i)] \frac{X \text{ ConstrXiX}}{X r_i} \\ D \quad \Gamma \quad (\text{IndX})(Qf_0 \cdots f_m \mu) : (QW)\mu \end{array}$$

当然, 正如之前的例子, 最后一句话也可以写成

$$\begin{aligned} \text{IndX} : \Pi(Z : \Pi(a : A)(Xa) \rightarrow \text{Type}_0) & ([F_{X|Q}^d(r_0, C_0)] \frac{X \text{ ConstrXiX}}{X r_0} \rightarrow \cdots \\ & \rightarrow [F_{X|Q}^d(r_m, C_m)] \frac{X \text{ ConstrXiX}}{X r_m} \rightarrow ZW) \end{aligned}$$

CIC: Compu

Definition 1.5.5 For any $Q, f_i, M_i, A_i \in F$, $X, i \in C$ and $X, a_i, \mu \in V$,

$$(\text{IndX})(Qf_0 \cdots f_m(\text{ConstrXi}M)) \rightarrow_X [F_{X|\lambda(a:A)\lambda(\mu:(Xa))}^e \text{IndX}Qf_0 \cdots f_m \mu(f_i, C_i) \frac{X}{X}] M$$

and Compatible $\rightarrow_X$.